**Solutions to Question 1:**

**(a)** In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost at each hour $\lambda_t$ yielding:

$$\frac{\partial C_{1,t}(P_{1,t})}{\partial P_{1,t}} = 1 + 0.1 * P_{1,t} = \lambda_t \rightarrow P_{1,t} = \frac{\lambda_t - 1}{0.01}$$

(1.1)

$$\frac{\partial C_{2,t}(P_{2,t})}{\partial P_2} = 6 + 0.5 * P_{2,t} = \lambda_t \rightarrow P_{2,t} = \frac{\lambda_t - 6}{0.5}$$

(1.2)

$$\frac{\partial C_{3,t}(P_{3,t})}{\partial P_{3,t}} = 15 + 1.1 * P_{3,t} = \lambda_t \rightarrow P_{3,t} = \frac{\lambda_t - 15}{1.1}$$

(1.3)

The satisfaction of the system demand implies that:

$$P_{1,t} + P_{2,t} + P_{3,t} = D_t$$

(1.4)

Substituting (1.1)-(1.3) in (1.4) gives for the system marginal cost:

$$\frac{\lambda_t - 1}{0.1} + \frac{\lambda_t - 6}{0.5} + \frac{\lambda_t - 15}{1.1} = D_t \rightarrow \lambda_1 = 18.2535 \left[\frac{£}{MWh}\right] \text{ and } \lambda_2 = 64.7324 \left[\frac{£}{MWh}\right] \quad (1.5)$$

Hour 2 is characterised by higher system marginal cost, since it exhibits a higher level of demand.

Substituting (1.5) in (1.1)-(1.3) yields for the optimal production of each generator:

$$P_{1,1} = 172.5352\ [MW] \text{ and } P_{1,2} = 637.3239\ [MW]$$

(1.6)

$$P_{2,1} = 24.5070\ [MW] \text{ and } P_{2,2} = 117.4648\ [MW]$$

(1.7)

$$P_{3,1} = 2.9577\ [MW] \text{ and } P_{3,2} = 45.2113\ [MW]$$

(1.8)

Substituting (1.6)-(1.8) in the cost functions of the generators yields for the total cost of operating the system over the 2-hour period:"

$$C_1 = \sum_t C_{1,t}(P_{1,t}) = 24007.3769\ [£]$$

(1.9)

$$C_2 = \sum_t C_{2,t}(P_{2,t}) = 5651.4739\ [£] \quad (1.10)$$

$$C_3 = \sum_t C_{3,t}(P_{3,t}) = 2851.5791\ [£] \quad (1.11)$$

$$C_{tot} = \sum_i \sum_t C_{i,t}(P_{i,t}) = 32510.4225\ [£]$$

(1.12)

Answers

The total profit for each generator $\Omega_i$ is given by the difference between its respective revenue and cost:

$\Omega_1 = \sum_t \left[\lambda_t P_{1,t} - C_{1,t}\left(P_{1,t}\right)\right] = 20397.5104$ [£]

(1.13)

$\Omega_2 = \sum_t \left[\lambda_t P_{2,t} - C_{2,t}\left(P_{2,t}\right)\right] = 2399.6429$ [£]

(1.14)

$\Omega_3 = \sum_t \left[\lambda_t P_{3,t} - C_{3,t}\left(P_{3,t}\right)\right] = 129.0438$ [£]

(1.15)

**(b)** At hour 1 when demand is low, given that $P_{3,1} < P_3^{min}$ the solution calculated in **(a)** is not feasible when taking into account the given output limits of the generators. The optimal solution for hour 1 can now be calculated by fixing the output of generator 3 to:

$P'_{3,1} = P_3^{min} = 20\ [MW]$

(1.16)

and satisfying the remaining demand:

$D'_1 = D_1 - P'_{3,1} = 180\ [MW]$

(1.17)

through the remaining generators 1 and 2. Following the same method in **(a)**, we get:

$\frac{\lambda'_1 - 1}{0.1} + \frac{\lambda'_1 - 6}{0.5} = 180 \rightarrow \lambda'_1 = 16.8333\ [\frac{£}{MWh}]$

(1.18)

Substituting (1.18) in (1.1)-(1.2) yields for the optimal production of each generator:

$P'_{1,1} = 158.3333\ [MW]$

(1.19)

$P'_{2,1} = 21.6667\ [MW]$

(1.20)

Similarly, at hour 2 when demand is high, given that $P_{2,2} > P_2^{max}$ the solution calculated in **(a)** is not feasible when taking into account the given output limits of the generators. The optimal solution for hour 2 can be calculated by fixing the output of generator 2 to:

$P'_{2,2} = P_2^{max} = 110\ [MW]$

(1.21)

and satisfying the remaining demand:

$D'_2 = D_2 - P'_{2,2} = 690\ [MW]$

(1.22)

Answers

through the remaining generators 1 and 3. Following the same method in **(a)**, we get:

$$\frac{\lambda_2'-1}{0.1}+\frac{\lambda_2'-15}{1.1}=690 \rightarrow \lambda_2' = 65.4167 \left[\frac{£}{MWh}\right]$$

(1.23)

Substituting (1.23) in (1.1) and (1.3) yields for the optimal production of each generator:

$$P_{1,2}' = 644.1667 \, [MW]$$

(1.24)

$$P_{3,2}' = 45.8333 \, [MW]$$

(1.25)

Substituting (1.15), (1.18)-(1.19), (1.20), and (1.23)-(1.24) in the cost functions of the generators yields for the total cost of operating the system:

$$C_1 = \sum_t C_{1,t}\left(P_{1,t}\right) = 24203.5069 \, [£]$$

(1.26)

$$C_2 = \sum_t C_{2,t}\left(P_{2,t}\right) = 5132.3611 \, [£] \quad (1.27)$$

$$C_3 = \sum_t C_{3,t}\left(P_{3,t}\right) = 3362.8819 \, [£] \quad (1.28)$$

$$C_{tot}' = \sum_i \sum_t C_{i,t}\left(P_{i,t}'\right) = 32698.7500 \, [£]$$

(1.29)

The profit for each generator is given by the difference between its respective revenue and cost:

$$\Omega_1' = \sum_t \left[\lambda_t' P_{1,t}' - C_{1,t}\left(P_{1,t}'\right)\right] = 20601.0069 \, [£]$$

(1.30)

$$\Omega_2' = \sum_t \left[\lambda_t' P_{2,t}' - C_{2,t}\left(P_{2,t}'\right)\right] = 2428.1944 \, [£]$$

(1.31)

$$\Omega_3' = \sum_t \left[\lambda_t' P_{3,t}' - C_{3,t}\left(P_{3,t}'\right)\right] = -27.9514 \, [£]$$

(1.32)

**(c)** The negative profit of generator 3 means that the operating cost of the latter is larger than its revenue in the optimal solution and thus it experiences economic losses at the optimal solution (does not recover its operating costs). This is happening due to the combination of the high variable costs of generator 3 and its high minimum output limit with respect to the initial unconstrained dispatch calculated in **(c)**. Since the generators are free to participate in the market based on their profitability, generator 3 can avoid the negative profits (economic losses) by decommitting from the market.

Answers

Solutions to Question 2:

**(a)** A brief discussion of the following points is expected:
1) There tend to be fairly large participants able to influence the energy price by their actions.
2) The low elasticity of demand facilitates the abuse of market power.
3) Locational effects could prevent the free flow of the commodity.
4) Electricity networks are natural monopolies.

**(b)** In the Cournot model of competition, the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column corresponds to a given production by $A$ and the cells in a row corresponds to a given production by company $B$.
Each cell contains four pieces of information arranged in the following format:

| | |
|---|---|
| $D$ | $\Omega_A$ |
| $\Omega_B$ | $\pi$ |

where:
$\pi$: price $\left[\frac{£}{MWh}\right]$
$D$: demand $[MW]$
$\Omega_A$: profit made by company $A$ $[\frac{£}{h}]$
$\Omega_B$: profit made by company $B$ $[\frac{£}{h}]$
Given the productions $P_A$ and $P_B$ of the two companies, the other quantities are calculated as follows:
$D = P_A + P_B$
(2.1)
$\pi = 250 - 2 * D$
(2.2)
$\Omega_A = P_A * (\pi - 75)$
(2.3)
$\Omega_B = P_B * (\pi - 62)$
(2.4)
Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below. A Nash equilibrium implies that given the production level of either of the two companies in this state, the other company cannot

ANSWERS

achieve a higher profit by allowing a production level different than the one at the equilibrium. It can thus be observed form the table below that the cell corresponding to $P_A = 30MW$ and $P_B = 30MW$ ; $P_A = 25MW$ and $P_B = 35MW$ are two Nash equilibria.

| $P_B/P_A$ | 20 | | 25 | | 30 | | 35 | |
|---|---|---|---|---|---|---|---|---|
| 25 | 45 | 1700 | 50 | 1875 | 55 | 1950 | 60 | 1925 |
| | 2450 | 160 | 2200 | 150 | 1950 | 140 | 1700 | 130 |
| 30 | 50 | 1500 | 55 | 1625 | 60 | 1650 | 65 | 1575 |
| | 2640 | 150 | 2340 | 140 | 2040 | 130 | 1740 | 120 |
| 35 | 55 | 1300 | 60 | 1375 | 65 | 1350 | 70 | 1225 |
| | 2730 | 140 | 2380 | 130 | 2030 | 120 | 1680 | 110 |
| 40 | 60 | 1100 | 65 | 1125 | 70 | 1050 | 75 | 875 |
| | 2720 | 130 | 2320 | 120 | 1920 | 110 | 1520 | 100 |

**(c)** The cell corresponding to $P_A = 20MW$ and $P_B = 25MW$ involves higher profits than the two cells corresponding to the Nash Equilibria: $P_A = 30MW$ and $P_B = 30MW$; $P_A = 25MW$ and $P_B = 35MW$ (identified as equilibria in **(b)**). However, this cell does not satisfy the properties of NE since: a) given that $P_A = 20MW$, company $B$ can achieve higher profits by producing $P_B = 35MW$ ($\Omega_B = 2{,}730\,£/h$) than $P_B = 25MW$ ($\Omega_B = 2{,}450\,£/h$) and also b) given that $P_B = 25MW$, company $A$ can achieve higher profits by producing $P_A = 30MW$ ( $\Omega_A = 1{,}950\,£/h$ ) than $P_A = 20MW$ ( $\Omega_A = 1{,}700\,£/h$).

**(d)** Each generating company is trying to maximize its profit which is equal to the difference between its revenue and its cost.

$\Omega_A = P_A * \pi(D) - C_A(P_A)$

(2.5)

$\Omega_B = P_B * \pi(D) - C_B(P_B)$

(2.6)

However, we cannot treat these two maximizations as separate optimization problems since they are links through the demand:

$D = P_A + P_B$

(2.7)

$\pi = 250 - 2 * D$

(2.8)

Answers

We can around this difficulty by expressing the price as a function of the production of the two generators:

$\pi = 250 - 2 * (P_A + P_B)$

(2.9)

The profit of the two companies than be expressed as:

$\Omega_A = P_A * (250 - 2 * P_A - 2 * P_B) - 75 * P_A = -2 * P_A^2 + (175 - 2 * P_B) * P_A$ (2.10)

$\Omega_B = P_B * (250 - 2 * P_A - 2 * P_B) - 62 * P_B = -2 * P_B^2 + (188 - 2 * P_A) * P_B$ (2.11)

The optimality condition for the maximization problem of company $A$ is:

$\frac{\partial \Omega_A}{\partial P_A} = -4 * P_A - 2 * P_B + 175 = 0$

(2.12)

The optimality condition for the maximization problem of company $B$ is:

$\frac{\partial \Omega_B}{\partial P_B} = -4 * P_B - 2 * P_A + 188 = 0$

(2.13)

(We only consider the partial derivative of the profit of a generator with respect to its own output because that is the only variable over which it has control)

Putting these optimality conditions in matrix form we get:

$$\begin{bmatrix} -4 & -2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} P_A \\ P_B \end{bmatrix} = \begin{bmatrix} -175 \\ -188 \end{bmatrix}$$

(2.14)

Solving this linear system we get the following productions for the NE:

$P_A = 27[MW]$

(2.15)

$P_B = 33.5[MW]$

(2.16)

Form which we can easily find:

$\pi = 129 \left[\frac{£}{MWh}\right], D = 60.5\ [MW], \Omega_A = 1{,}485[\frac{£}{h}], \Omega_B = 2{,}244.5[\frac{£}{h}]$

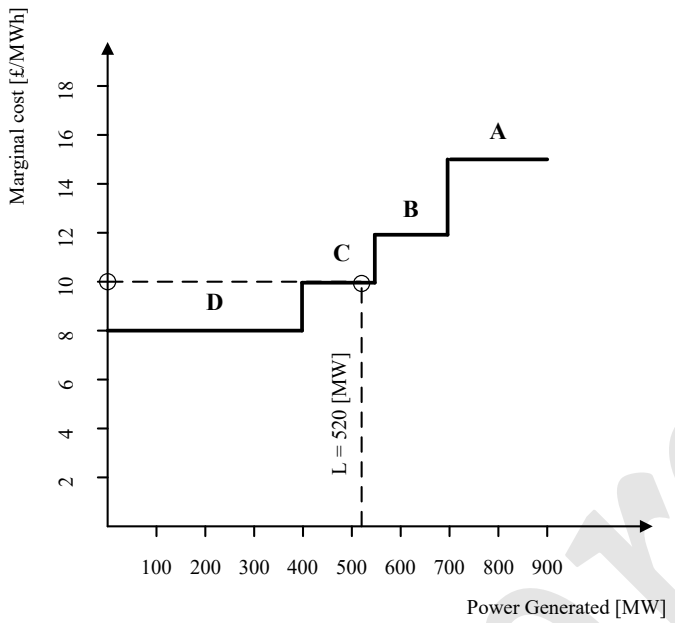
(2.17)

Answers

Answers

**Solutions to Question 3**

**(a)** Since there are no transmission constraints, the output of all the generators can be stacked in order of marginal cost as shown in the following figure:




Using the figure, we can see that for a system load of $L = 400 + 40 + 80 = 520MW$ the marginal cost (and hence the nodal price) is:

$\pi = 10\ [\frac{£}{MWh}]$

(3.1)

Furthermore, the units are dispatched as follows:

$P_D = 400\ [MW]$

(3.2)

$P_C = 120\ [MW]$

(3.3)

$P_A = P_B = 0[MW]$

(3.4)

The line flow problem can be solved using the superposition principle and also directly. To this effect, we write the power balance equation at two buses and KVL around the loop.

Bus 1: $P_A - 400 = F_{12} + F_{13}$

(3.5)

Bus 2: $P_B - 80 = -F_{12} + F_{23}$

(3.6)

Bus 3: $P_C + P_D - 40 = -F_{13} - F_{23}$

(3.7)

Loop equation: $0.2F_{12} + 0.3F_{23} - 0.3F_{13} = 0$

(3.8)

Putting these equations in the matrix form gives:

ANSWERS

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_B - 80 \\ P_C + P_D - 40 \\ 0 \end{bmatrix}$$

(3.9)

Substituting $P_A = 0\ [MW]$, $P_B = 0[MW]$, $P_C = 120[MW]$ and $P_D = 400[MW]$ in (3.9), we get:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} -80 \\ 480 \\ 0 \end{bmatrix}$$

(3.10)

Solving these equations, we get:

$F_{12} = -120\ [MW]$

(3.11)

$F_{13} = -280\ [MW]$

(3.12)

$F_{23} = -200[MW]$

(3.13)

**(b)** From part b), the flow on line 1-3 exceeds its maximum capacity by $30[MW]$.

**(i)** The first method consists in increasing the output of generator A and decreasing by the same amount the output of generator C. Decreasing the output of generator D is not desirable as it is cheaper than generator C. To calculate how big this increase should be to remove the violation of the flow limit on line 3-1, consider an injection of $+1[MW]$ at bus 1 and an injection of $-1[MW]$ at bus 3. This pair of injection causes a flow in the network that divides itself as follows:

$\frac{0.3}{(0.2+0.3)+0.3} \cdot 1 = 0.375[MW]$ along the path 1-2-3

(3.14)

$\frac{(0.2+0.3)}{(0.2+0.3)+0.3} \cdot 1 = 0.625[MW]$ along the path 1-3

(3.15)

Since we use a linear (dc) model, we can say that to remove $30[MW]$ overload on line 3-1, we therefore need to increase the output of generator A by:

$\frac{30}{0.625} = 48\ [MW]$

(3.16)

The constrained dispatch is then:

$P_A = 48\ [MW]$

(3.17)

$P_B = 0\ [MW]$

(3.18)

Answers

$P_C = 72[MW]$

(3.19)

$P_D = 400[MW]$

(3.20)

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_A - 400 \\ P_C + P_D - 40 \\ 0 \end{bmatrix}$$

(3.21)

We get:

$F_{12} = -102\ [MW]$

(3.22)

$F_{13} = -250\ [MW]$

(3.23)

$F_{23} = -182[MW]$

(3.24)

Then, we can see that this dispatch does not cause a violation of the line flow constrains on any other line.

The cost of this dispatch is:

$C_{Total} = 48 \cdot 15 + 72 \cdot 10 + 400 \cdot 8 = 4640[£]$

(3.25)

**(ii)** The other method to remove the constraint violation consists in increasing the output of generator B and decreasing the output of generator C by the same amount. To calculate how big this increase should be to remove the violation of the flow on line 3-1, consider an injection of $+1[MW]$ at bus 2 and an injection of $-1[MW]$ at bus 3. This pair of injection causes a flow in the network that divides itself as follows:

$\frac{0.3}{(0.2+0.3)+0.3} \cdot 1 = 0.375[MW]$ along the path 2-1-3

(3.26)

$\frac{(0.2+0.3)}{(0.2+0.3)+0.3} \cdot 1 = 0.625[MW]$ along the path 2-3

(3.27)

Since we use a linear (dc) model, we can say that to remove $30[MW]$ overload on line 3-1, we therefore need to increase the output of generator B by:

$\frac{30}{0.375} = 80\ [MW]$

(3.28)

The constrained dispatch is then:

$P_A = 0\ [MW]$

(3.29)

Answers

$P_B = 80\ [MW]$

(3.30)

$P_C = 40[MW]$

(3.31)

$P_D = 400[MW]$

(3.32)

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_B - 80 \\ P_C + P_D - 40 \\ 0 \end{bmatrix}$$

(3.33)

We get:

$F_{12} = -150\ [MW]$

(3.34)

$F_{13} = -250\ [MW]$

(3.35)

$F_{23} = -150[MW]$

(3.36)

Once again, this redispatch does not cause a violation of the line flow constrains on any other line. The cost of this constrained dispatch is:

$C_{Total} = 80 \cdot 12 + 40 \cdot 10 + 400 \cdot 8 = 4560[£]$

(3.37)

Even though it re-dispatches a larger amount of $MW$, the second constrained dispatch is preferable to the first one because its cost is smaller. It is thus the optimal constrained dispatch.

**(c)** The nodal price at each bus is given by the cost of one additional $MW$ of load at each node. Therefore, the price at bus 3 is 10 $\left[\frac{£}{MWh}\right]$ because the next $MW$ of load would be generated locally by generator C because it is the cheapest generator not operating at its upper limit. An additional $MW$ of load at node 2 would have to be produced by generator B. Producing it with generator C would cause a violation of the line flow constraint on line 3-1. Producing it with generator A would be more expensive than with generator B. The price at node 2 is therefore 12 $\left[\frac{£}{MWh}\right]$. An additional $MW$ of load at bus 1 requires a redispatch of generators B and C to minimize the cost increase while maintaining the flow on line 3-1 within limits.

Extracting an additional $1[MW]$ at bus 1 and generating it at bus 3 causes the following change in the flow on line 1-3:

Answers

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \Longrightarrow \Delta F_{31} = 0.625[MW]$$

(3.38)

Similarly, extracting an additional $1[MW]$ at bus 1 and generating it at bus 2 causes the following change in the flow on line 1-3:

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0.2 & -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \Longrightarrow \Delta F_{31} = 0.250[MW]$$

(3.39)

Therefore, if we want that the flow on line 3-1 remains unchanged (because it is already at its limit), we must change the productions by generators C and B in such a way that:

$$\Delta F_{31} = 0 = 0.625 \cdot \Delta P_C + 0.250 \cdot \Delta P_B$$

(3.40)

At the same time, since we are increasing the load by $1[MW]$, we must also have:

$$\Delta P_B + \Delta P_C = 1$$

(3.41)

Solving the system consisting of the previous two equations we get:

$$\Delta P_C = -0.667[MW]$$

(3.42)

$$\Delta P_B = 1.667[MW]$$

(3.43)

To supply an additional $MW$ of load at bus 1 without violating the network constraints, we must therefore increase the output of generator B and decrease the output of generator C. The nodal price at bus 1 is thus given by a linear combination of the marginal cost of production of these two generators.

$$\pi_1 = -0.667 \cdot 10 + 1.667 \cdot 12 = 13.33[\frac{£}{MWh}]$$

(3.44)

Finally:

$$\pi_1 = 13.33[\frac{£}{MWh}]$$

(3.45)

$$\pi_2 = 12[\frac{£}{MWh}]$$

(3.46)

$$\pi_3 = 10[\frac{£}{MWh}]$$

(3.47)

Answers

**Solution to Question 4**

**(a)-(i)** We neglect the transmission line constraint.

**Winter**

In area A, the demand can be only supplied by generator A, so the dispatch of generator A is

$$P_A = D_A = 1{,}000MW \quad (4.1)$$

The electricity price in area A is

$$\lambda_A = MC_A = 1 + 0.002 \cdot P_A = 3\ £/\text{MWh} \quad (4.2)$$

In area B, the demand can be only supplied by generator B, so the dispatch of generator B is

$$P_B = D_B = 2{,}500MW \quad (4.3)$$

The electricity price in area B is

$$\lambda_B = MC_B = 2 + 0.004 \cdot P_B = 12\ £/\text{MWh} \quad (4.4)$$

**Summer**

In area A, the demand can be only supplied by generator A, so the dispatch of generator A is

$$P_A = D_A = 500MW \quad (4.5)$$

The electricity price in area A is

$$\lambda_A = MC_A = 1 + 0.002 \cdot P_A = 2\ £/\text{MWh} \quad (4.6)$$

In area B, the demand can be only supplied by generator B, so the dispatch of generator B is

$$P_B = D_B = 1{,}500MW \quad (4.7)$$

The electricity price in area B is

$$\lambda_B = MC_B = 2 + 0.004 \cdot P_B = 8\ £/\text{MWh} \quad (4.8)$$

**(a)-(ii)** We impose the unlimited transmission line constraint.

**Winter**

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ $subject\ to$ $P_A + P_B = 3500$

Answers

$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 3500)$

(4.9)

$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002}$

(4.10)

$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004}$

(4.11)

$P_A + P_B = 3500 \rightarrow \lambda = 6 \frac{£}{MWh} = MC_A = MC_B, P_A = 2{,}500MW, P_B = 1{,}000MW$

(4.12)

**Summer**

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ *subject to* $P_A + P_B = 2000$

$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 2000)$

(4.13)

$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002}$

(4.14)

$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004}$

(4.15)

$P_A + P_B = 2000 \rightarrow \lambda = 4 \frac{£}{MWh} = MC_A = MC_B, P_A = 1{,}500MW, P_B = 500MW$ (4.16)

**(b)** We take into account the transmission line constraint of 800MW.

**Winter**

The original power flow is 1500MW, now there will be 700MW reduction if the maximum power flow is 800MW.

The optimal dispatch becomes:

$P_A = 2500MW - 700MW = 1{,}800MW$ (4.17)

$P_B = 1000MW + 700MW = 1{,}700MW$ (4.18)

and the prices are:

$\lambda_A = MC_A = 1 + 0.002 \cdot P_A = 4.6$ £/MWh

(4.19)

$\lambda_B = MC_B = 2 + 0.004 \cdot P_B = 8.8$ £/MWh

(4.20)

The hourly generation costs are

$C_A = 1 \cdot 1800 + 0.001 \cdot 1800^2 = 5{,}040$ £/h

(4.21)

Answers

$C_B = 2 \cdot 1700 + 0.002 \cdot 1700^2 = 9{,}180$ £/h
(4.22)

**Summer**

The original power flow is 1000MW, now there will be 200MW reduction if the maximum power flow is 800MW.

The optimal dispatch becomes:

$P_A = 1500MW - 200MW = 1{,}300MW$ (4.23)

$P_B = 500MW + 200MW = 700MW$ (4.24)

and the prices are:

$\lambda_A = MC_A = 1 + 0.002 \cdot P_A = 3.6$ £/MWh
(4.25)

$\lambda_B = MC_B = 2 + 0.004 \cdot P_B = 4.8$ £/MWh
(4.26)

The hourly generation costs are

$C_A = 1 \cdot 1300 + 0.001 \cdot 1300^2 = 2{,}990$ £/h
(4.27)

$C_B = 2 \cdot 700 + 0.002 \cdot 700^2 = 2{,}380$ £/h
(4.28)

**(c) (i)** We take into account the transmission line constraint of 800MW.

The annual generation costs are:

$C_A^{winter} = 5{,}040 \frac{£}{h} \cdot 2700h = 13{,}608{,}000$ £ (4.29)

$C_B^{winter} = 9{,}180 \frac{£}{h} \cdot 2700h = 24{,}786{,}000$ £ (4.30)

$C_A^{summer} = 2{,}990 \frac{£}{h} \cdot 6060h = 18{,}119{,}400$ £ (4.31)

$C_B^{summer} = 2{,}380 \frac{£}{h} \cdot 6060h = 14{,}442{,}800$ £ (4.32)

$C_{A+B}^{annual} = 70{,}936{,}200$ £ (4.33)

**(c) (ii)**

In winter, the electricity price in area A is 4.6 £/MWh and the electricity price in area B is 8.8 £/MWh.

The winter generation revenue are:

$1800MW \cdot 2700h \cdot 4.6 \frac{£}{MWh} + 1700MW \cdot 2700h \cdot 8.8 \frac{£}{MWh} = 62{,}748{,}000£$ (4.34)

The winter demand charges are:

$1000MW \cdot 2700h \cdot 4.6 \frac{£}{MWh} + 2500MW \cdot 2700h \cdot 8.8 \frac{£}{MWh} = 71{,}820{,}000£$ (4.35)

Answers

In summer, the electricity price in area A is 3.6 £/MWh and the electricity price in area B is 4.8 £/MWh.

The summer generation revenue are:

$$1300MW \cdot 6060h \cdot 3.6\frac{£}{MWh} + 700MW \cdot 6060h \cdot 4.8\frac{£}{MWh} = 48{,}722{,}400£ \quad (4.36)$$

The summer demand charges are:

$$500MW \cdot 6060h \cdot 3.6\frac{£}{MWh} + 1500MW \cdot 6060h \cdot 4.8\frac{£}{MWh} = 54{,}540{,}000£ \quad (4.37)$$

**So, the annual generation revenue and demand charge are:**

$$R_{winter+summer}^{annual} = 62{,}748{,}000 + 48{,}722{,}400 = 111{,}470{,}400\ £ \quad (4.38)$$

$$C_{winter+summer}^{annual} = 71{,}820{,}000 + 54{,}540{,}000 = 126{,}360{,}000\ £ \quad (4.39)$$

**(c) (iii)** The TransCo's total annual revenue is

$$R_{Tra} = 800 \cdot (8.8 - 4.6) \cdot 2700 + 800 \cdot (4.8 - 3.6) \cdot 6060 = 14{,}889{,}600£ \quad (4.40)$$

We can see that it is equal to the difference between annual demand charges and annual generation revenues.

Answers